

# Certified Lattice Data and a Within-Family Selector for the Cooper $s_7/s_{10}$ $K3$ Families

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*Stream 2 — Selection & Geometry (SocrateAI-Scientific-Agora-K3-DarkMatter)*

2026-09-29 (preprint; not reviewed)

## What is not claimed

This is a mathematics note about lattice arithmetic on two families of  $K3$  surfaces. It makes no physical claim of any kind: no dark-matter identification, no dark-energy statement, no observable. The programme’s flux/tadpole route is blocked pending a specified threefold base (F5b) and a selection among the surfaces discussed here maps to no measurement. These renunciations are load-bearing, not boilerplate; see section 1.

## Abstract

We report a session’s results on the Cooper  $s_7$  and  $s_{10}$  Picard–Fuchs families, all of it lattice mathematics with an explicit epistemic tier on every claim. Six points of the  $\rho = 20$  locus of each family’s transcendental lattice are exhibited as exact vectors with saturated orthogonal complements, reduced binary forms, and frame indices; the lattice arithmetic of all twelve is kernel-checked in Lean 4, three of the six *independently* by two separate developments, which we re-verify ourselves before citing either. A within-family selector — minimal discriminant of the transcendental lattice — is adopted by an explicit ruling among two computed alternatives, chosen because the rejected alternative is not unique in one family. An explicit  $M_7$ -polarized model realizes the  $s_7$  family as an Inose surface; its elliptic fibration, taken from a sourced literature equation and re-derived here, has the two singular loci of the Picard–Fuchs operator indistinguishable at the level of discriminant-root orders, which answers a standing question about one of them in the negative. Stage-2 monodromy of the order-two operator underlying each family is certified by rigorous ball arithmetic, replacing an earlier numerical-tolerance heuristic with enclosures whose diameters are reported exactly. We record, in band, five errors made and caught during the same session, because the standing discipline of this programme requires retractions to be machine-visible rather than a footnote.

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## 1 Scope and what is not claimed

### What is not claimed

No claim in this note is physical. The  $\rho = 20$  locus is a set of lattice points on a modular curve; it maps to no observable and licenses no dark-sector identification. The discriminant-floor selector of section 3 is a naming convention for which candidate a certificate calls “distinguished” within a family, adopted by an explicit ruling, and it does not rank one Cooper family against the other. No Kodaira fibre type is attached to any point of either family’s Picard–Fuchs operator: that reading is a category error for this class of family (elliptic points of a modular curve are not Kodaira degenerations), and the one place an explicit Weierstrass model is computed (section 4.1) we report discriminant-root *orders* only, withholding the label pending a separate ruling. The  $s_{10}$  family’s transcendental lattice remains a DRAFT certificate; every  $s_{10}$  result here is advisory and does not depend on, or promote, that certificate’s status.

## 2 The six $\rho = 20$ locus rows, kernel-checked

Each register family ( $s_7$  at level  $N = 7$ ,  $s_{10}$  at level  $N = 10$ ) carries a rank-three lattice  $U \oplus \langle 2N \rangle$  with quadratic form  $v^2 = 2xy + 2Nz^2$  on coordinates  $v = (x, y, z)$ . Three points per family are singled out by the vanishing locus of the family’s order-three Picard–Fuchs operator  $L_3$ : the two finite singular points and the point at infinity. At each such point the transcendental lattice, rank 19 generically, drops to rank 2 and becomes positive definite —  $\rho$  jumps to 20. Table 1 lists all six, as exact vectors with their norm, the exact orthogonal complement (its Gram matrix, reduced to a binary quadratic form of discriminant  $D$ ), and the Lean kernel status.

| Family                | $z$      | $v$           | $v^2$ | $D$ | $T$ (reduced) | Tier / sources      |
|-----------------------|----------|---------------|-------|-----|---------------|---------------------|
| cooper_s7             | −1       | (2, −4, 1)    | −2    | −7  | [2, 1; 1, 4]  | Tier A, 2 source(s) |
| cooper_s7             | 1/27     | (1, −1, 0)    | −2    | −28 | [2, 0; 0, 14] | Tier A, 2 source(s) |
| cooper_s7             | $\infty$ | (14, −14, −5) | −42   | −3  | [2, 1; 1, 2]  | Tier A, 2 source(s) |
| cooper_s10 (ADVISORY) | −1/4     | (2, −6, 1)    | −4    | −20 | [4, 2; 2, 6]  | Tier A, 1 source(s) |
| cooper_s10 (ADVISORY) | 1/16     | (1, −1, 0)    | −2    | −40 | [2, 0; 0, 20] | Tier A, 1 source(s) |
| cooper_s10 (ADVISORY) | $\infty$ | (10, −10, −3) | −20   | −4  | [2, 0; 0, 2]  | Tier A, 2 source(s) |

Table 1: The six locus rows. **Tier A — Lean kernel** for the lattice arithmetic (norm, saturated complement, Gram, reduced form, frame determinant) of every row; three rows carry two independent kernel developments. **Tier B — derived, not measured**: the numeric recognition of  $z$ , and the identification of the orthogonal complement with the family’s transcendental lattice  $T_X$  (Dolgachev 1996 §7, Doran 1998 Thm. 5.13, **Tier L — literature**). The index step  $|\det(\text{frame})| = |v^2|/\text{div}(v)$  is checked per row, never proved in general, in either kernel development.

### Headline result

For the two rows at  $z = 1/27$  and  $z = -1$  (the  $s_7$  family) and, independently added this session, both families'  $z = \infty$  rows, the lattice half is kernel-checked *twice*, by two developments written without reading each other's source, then read at the statement level and re-gated by us in each producer's own build tree before either is cited.

*Remark 2.1.* The two independent developments state the norm, the exact orthogonal complement as a set (an if-and-only-if, not merely two orthogonal vectors), the complement's Gram matrix, its Gauss-reduced form, and the frame determinant, each as a standalone kernel-checked declaration, over the standard axiom set `{propext, Classical.choice, Quot.sound}` and nothing else. Producer and verifier are different in every case reported here: we re-ran the build, the sorry grep, the axiom audit, the statement lock, and `#print axioms` ourselves, in the producer's own worktree, before writing any lattice tier into a certificate.

## 3 An adopted selector, and the alternative it rejected

Six candidates are too many to call “the” surface of a family without saying which quantity is being extremized. Two computed candidates were compared:

- **SEL-D**: minimal  $|\text{disc } T|$  within a family, verified unique not merely by class number (which counts reduced forms) but by the count of points *on the modular curve* itself.
- **SEL-N**: minimal  $|v^2|$  within a family — the  $-2$  floor achieved by an  $N$ -uniform Lean theorem in both kernel developments.

Table 2 gives both, exactly, per family.

| Family                             | SEL-D (min $ \text{disc } T $ ) | SEL-N (min $ v^2 $ )            | Agree? |
|------------------------------------|---------------------------------|---------------------------------|--------|
| <code>cooper_s7</code>             | $T = [2, 1; 1, 2], D = -3$      | tie: $D = -7, D = -28$ (2 rows) | No     |
| <code>cooper_s10</code> (ADVISORY) | $T = [2, 0; 0, 2], D = -4$      | $D = -40$ (unique)              | No     |

Table 2: SEL-D and SEL-N disagree in both families. SEL-N is *not unique* for  $s_7$ : two rows tie at the  $-2$  floor, corresponding to the two fixed points of the Atkin–Lehner involution  $W_7$  on the family's Hauptmodul coordinate.

**SEL-D was adopted.** The choice was put, with both alternatives computed and the disagreement stated, to the programme's designated decision authority, who selected SEL-D directly. This narrows an earlier, separately-ruled clause (“no minimum-discriminant rule”) to cross-family comparisons only; it does not rank  $s_7$  against  $s_{10}$ , and it does not promote  $s_{10}$ 's DRAFT lattice certificate. The selected surfaces —  $T = A_2$  ( $D = -3$ ) for  $s_7$ ,  $T = \langle 2 \rangle \oplus \langle 2 \rangle$  ( $D = -4$ ) for  $s_{10}$ , advisory — extremize the same quantity that a separate, pre-existing attractor-mechanism development in this programme's Lean corpus already names for the smallest admissible black-hole charge configuration; both are instances of the same discriminant-gap floor ( $4ac - b^2 \geq 3$ ), so the coincidence is forced by that shared floor and is stated once, here, as an observation and not as corroborating evidence for either result.

## 4 An explicit $M_7$ -polarized model

Restricting the Clingher–Doran–Lewis–Whitcher normal form for  $M$ -polarized  $K3$  surfaces to the  $M_7$ -polarized locus realizes the  $s_7$  family as an Inose surface  $X(a(z), b(z))$ , with  $a^3 = \pi(z)$ ,

$$b^2 = \pi(z) - \sigma(z) + 1 \text{ and}$$

$$\sigma(z) = J(\tau) + J(7\tau), \quad \pi(z) = J(\tau) J(7\tau),$$

where  $z$  is the family’s own  $\Gamma_0(7)^+$  Hauptmodul coordinate and  $J$  is Klein’s  $J$ -invariant, normalized  $J(i) = 1$ . Because  $X_0(7)^+$  has a single cusp,  $\sigma$  and  $\pi$  are Laurent polynomials in  $z$ ; both were solved from their  $q$ -expansions and verified against every held-out coefficient to order  $q^{128}$  (Tier A — Lean kernel for the finite-order verification; Tier B — derived, not measured for the identification of  $z$  with the Hauptmodul coordinate). Exactly,

$$\sigma(z)^2 - 4\pi(z) = \frac{-(z+1)(27z-1)(2z-1)^2(8z-1)^2(24z-1)^2(8z^2+16z-1)^2}{2985984 z^{14}}.$$

The *simple* zeros of this expression are exactly the two finite singular loci of  $L_3$  ( $z = -1$ ,  $D = -7$ ;  $z = 1/27$ ,  $D = -28$ ): the points where the two isogenous elliptic curves become isomorphic, which is where  $J(\tau) - J(7\tau)$  behaves like a square root, forced by the involution that swaps the two curves. The double zeros are the other self-7-isogenous CM points in the same window, each occurring exactly  $h(D)$  times.

#### 4.1 The fibration

A degree-two base change of a sourced elliptic pencil on  $\text{Km}(E_1 \times E_2)$  (Kuwata–Shioda, §5.3, Tier L — literature) gives the Inose fibration explicitly, with the transcription checked against the source’s own printed discriminant before any further computation. Table 3 gives the discriminant-root orders — never a Kodaira label, per section 1 — for the generic case, both  $s_7$  singular loci, and the two special values  $J = 0, 1$ .

| Case                             | Discriminant-root orders | Note            |
|----------------------------------|--------------------------|-----------------|
| Generic ( $E_1 \not\cong E_2$ )  | $\{10, 10, 1, 1, 1, 1\}$ | —               |
| cooper_s7, $z = -1$              | $\{10, 10, 2, 1, 1\}$    | $J = -125/64$   |
| cooper_s7, $z = 1/27$            | $\{10, 10, 2, 1, 1\}$    | $J = 614125/64$ |
| $J = 1$ ( $E_1 \cong E_2$ )      | $\{10, 10, 2, 2\}$       | —               |
| $J = 0$ ( $E_i \cong E_\omega$ ) | $\{10, 10, 4\}$          | —               |

Table 3: Discriminant-root orders of the Inose fibration. The mechanism behind the middle two rows: when the two elliptic curves become isomorphic, the discriminant’s constant term vanishes identically, and after the base change the two simple roots collapse onto the branch point, producing exactly one order-2 root there.

#### Headline result

The two singular loci of  $s_7$  are *indistinguishable* at the level of discriminant-root orders: both give  $\{10, 10, 2, 1, 1\}$ . A standing hypothesis in this programme — that the two loci differ in their fibre configuration — is answered in the negative by this computation, for this fibration. The second algebraic class responsible for the rank jump at either point must therefore be a Mordell–Weil section, not a reducible fibre.

## 5 Certified stage-two monodromy

The order-two Picard–Fuchs operator underlying each family’s  $\text{Sym}^2$  structure has monodromy that was previously recognized only by a numeric-tolerance heuristic (60-digit precision, a  $10^{-35}$  gate). This session replaced that heuristic with rigorous ball arithmetic (Arb, via `python-flint`):

Frobenius series tails and Taylor-continuation tails are bounded by majorant induction on the operator’s own recurrence, giving enclosures whose diameters certify that each recognized monodromy entry is the *unique* rational of bounded denominator inside its enclosure. Table 4 reports the diameters achieved; the certified matrices were then fed into the exact lattice pipeline and reproduce every field of the corresponding lattice certificate, closing the chain from certified numerics to exact lattice arithmetic.

| Family     | Clause     | Enclosure diameter     |
|------------|------------|------------------------|
| cooper_s7  | $z = -1$   | $1.6 \times 10^{-96}$  |
| cooper_s7  | $z = 1/27$ | $1.0 \times 10^{-115}$ |
| cooper_s10 | $z = -1/4$ | $4.2 \times 10^{-102}$ |
| cooper_s10 | $z = 1/16$ | $2.2 \times 10^{-115}$ |

Table 4: Enclosure diameters of the certified stage-two monodromy. All are many orders of magnitude below the denominator bound required for the recognized rational to be unique.

## 6 Corrections recorded in band

### Corrections recorded in this session

Five errors were made and caught within this session, listed because the standing discipline of this programme requires a retraction to be machine-visible, not a prose footnote.

1. An initial claim that SEL-N (section 3) was unique and level-uniform was false: two rows of the  $s_7$  family tie at the  $-2$  norm floor. Caught before any recommendation was written down.
2. An initial reading of a standing “no minimum-discriminant rule” as scoped to cross-family comparison only was a stretch of the text, not what it said; the question was put to the decision authority rather than resolved by inference.
3. A transcription of the Inose (0,0)/(1,0) labels for two special points was swapped; corrected after an independent Weierstrass-reduction computation disagreed with it.
4. Two software defects were found while extending the Lean-attestation machinery of section 2: a worktree-path resolver that only recorded one build tree per repository, so a second, later commit on a different branch of the same repository silently failed its own file-hash check; and a test control that matched a lattice row by vector coordinates alone, which collide across the two register families.
5. A version-control error landed a pull request containing only a brief document, omitting the actual code and data changes its own description claimed; caught by comparing a certificate’s hash on the merged branch against the local file, and corrected by a follow-up commit before any downstream party was told the (wrong) hash.

## 7 Reproducibility

Every number in tables 1 to 4 is generated directly from certificates in `data/certificates/` by `scripts/render_paper_tables.py`, which fails closed (`-check`) if any table fragment drifts from the certificates that produced it; nothing in this document’s tables is a typed literal. The checkers producing the underlying certificates each ship negative controls that must fail on a

tampered input before the corresponding positive result is trusted. A full account of every certificate, checker, and the exact gate output for both independent Lean developments of section 2 is in the repository’s decision and status records, referenced in section 8.

## Provenance

Generated-by: Claude (Sonnet 5), Stream 2 | Verified-by: `scripts/render_paper_tables.py -check`; every cited checker’s own controls | Reviewed-by: N (preprint; not reviewed).

## 8 References and source records

### References

- [1] I. Dolgachev, “Mirror symmetry for lattice polarized  $K3$  surfaces,” *arXiv:alg-geom/9502005* (1996).
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- [4] A. Clingher, C. F. Doran, J. Lewis, and U. Whitcher, “Normal forms,  $K3$  surface moduli, and modular parametrizations,” *arXiv:0712.1880*.
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- [7] G. Moore, “Arithmetic and Attractors,” *arXiv:hep-th/9807087* (1998).
- [8] SocrateAI Scientific Agora Collaboration, **SocrateAI-Scientific-Agora-K3-DarkMatter**, `K3_CRITERIA.md` and `TODO.md` at commit `d3a2a01` (2026-09-29); decision records `briefs/T0_DECISIONS_2026_09_28_STREAM2.md` (selector ruling), `briefs/STREAM2_AM8_SELECTOR_COMPARISON_2026_09_28.md` (comparison detail), `briefs/WP_GE10_INOSE_MODEL_M7_2026_09_27.md` (fibration derivation).